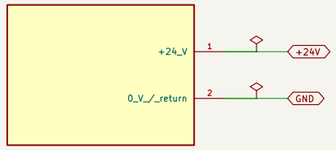
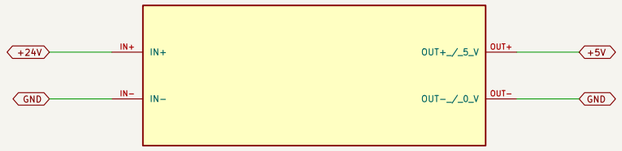
CONTROLLER AND POWER

J1
U1 barrel jack / 24 V



U1
QIQIAZI 5 V / 5 A



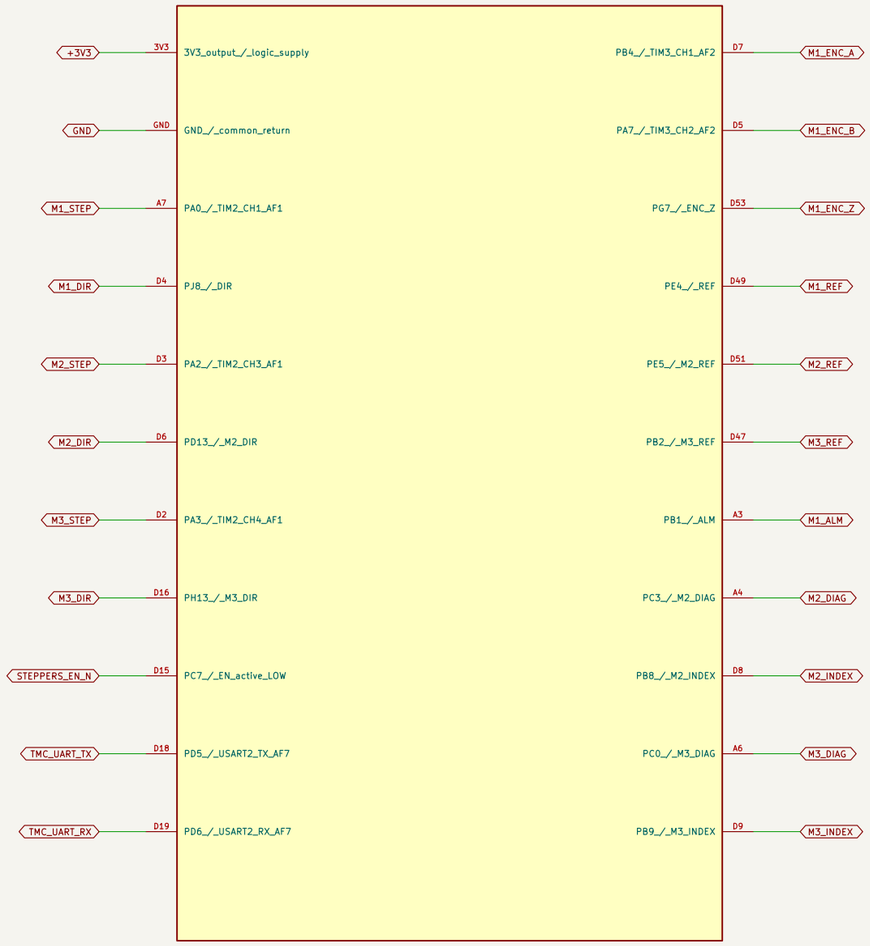
J1 is the existing barrel jack on buck U1.
J103 VIN pins feed J104.9/10 and both TMC power headers.
Dedicated underside supply/return branches; common 0 V.
Buck +5 V powers the DM542T signals and encoder circuit only.

GIGA POWER / LOGIC
Power the GIGA independently through USB−C.
GIGA 3V3 −> J101.8; GIGA GND −> common logic return.
Do not connect buck +5 V to GIGA 5V, VIN or USB.
No 24 V or 5 V on any GIGA signal input.
Use Arduino D/A labels, not connector ordinal numbers.
The GIGA has its own 16 MHz crystal; no ST−Link clock feed.

M4 MOTOR PERIPHERALS
TIM2 CH1/3/4: A7/PA0, D3/PA2, D2/PA3.
TIM3 encoder: D7/PB4, D5/PA7; index D53/PG7.
R7 biases shared enable HIGH (disabled) during reset.
R1/R2, R8 and R9 bias STEP LOW.
QSPI and filesystem storage are handled by the GIGA;
no external flash or SD wiring is part of this perfboard.

Shared names connect across sheets. A1 numbers are Arduino D/A/power labels; J101−J111/J113 are perfboard contacts.
REF inputs: internal 3.3 V pull−ups, both−edge EXTI; NC to GND. Released = LOW; pressed / unplugged = HIGH.
U1/A3/A4 mount at J103/J105/J106. Buck VIN distributes 24 V. TMC JP1.6/5: power headers; JP1.4−1: motor terminals.

A1
Arduino GIGA R1 WiFi / ABX00063



M1 − DM542T

Datei: sheets/M1_DM542T.kicad_sch

M2 and M3 − TMC2209

Datei: sheets/M2_M3_TMC2209.kicad_sch

M1 − encoder interface

Datei: sheets/M1_Encoder.kicad_sch

Perfboard cable headers

Datei: sheets/Perfboard_Connectors.kicad_sch

Reference limit switches

Datei: sheets/Reference_Switches.kicad_sch

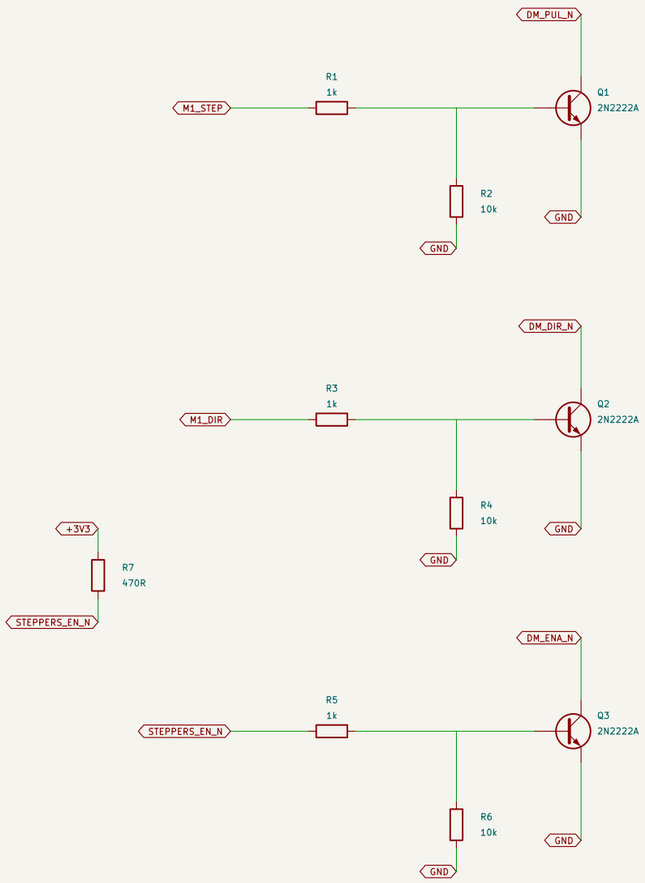
System wiring and linked perfboard assembly map | all voltages DC
ProtoMate Control
Sheet: /
File: ProtoMate.kicad_sch

Title: ProtoMate SCARA − Controller and power

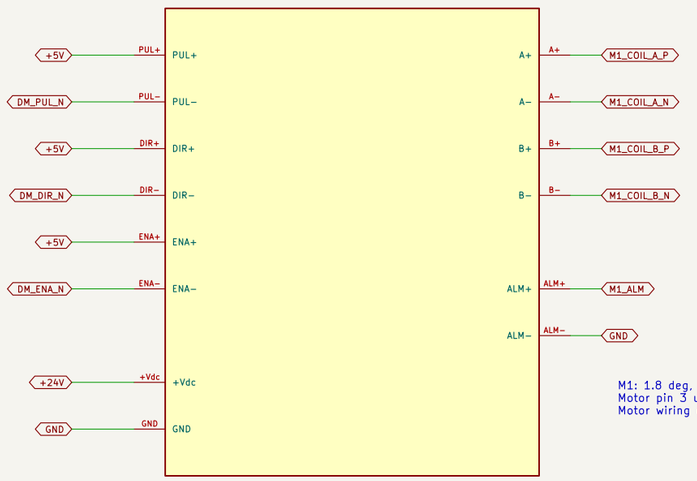
Size: A3	Date: 2026−10−15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 1/6

M1 – DM542T

3.3 V MCU –> 2N2222A current sinks –> 5 V optoisolated inputs

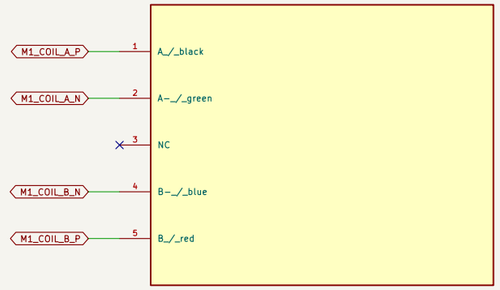


A2
DM542T V4.0



24 V power cable: J104.9 –> +Vdc; J104.10 –> GND.
Signal cable: J104. Motor winding cable stays on A2.

M1
PKP245D23A2–R2FL



M1: 1.8 deg, 200 full steps/rev, 2.3 A/phase.
Motor pin 3 unused. Encoder is on sheet 4.
Motor wiring colours refer to the motor drawing.

DM542T SETTINGS / INITIAL COMMISSIONING
S2 signal selector: 5 V.
1/16 step: SW5 ON, SW6 ON, SW7 OFF, SW8 ON.
3200 STEP pulses per motor revolution.
Conservative initial current: 2.03 A RMS / 2.84 A peak
SW1 ON, SW2 ON, SW3 OFF; SW4 OFF (idle reduction).
Check the actual V4.0 label before setting switches.



SHARED ENABLE AND TIMING
PC7 low: Q3 off, DM542T enabled, both TMC EN inputs low.
PC7 high: Q3 on, DM542T disabled, both TMC EN inputs high.
R7 pulls PC7 high during MCU reset; about 7 mA sink when enabled.
After enable, wait >=200 ms. DIR setup >=5 us; STEP high/low >=2.5 us.

Q1–Q3 show functional B/C/E pins, not package lead numbers.
Verify your 2N2222A manufacturer/package before assembly.
Fit each 10k between base and emitter. Keep PUL+/PUL– paired;
separate signal cables from motor power cables.

ALARM: ALM+ –> J104.7; ALM– –> J104.8 / GND.
J101.4 –> PB1 / A3; J101 GND returns shared.
R30 pulls ALM+ to 3.3 V (about 0.7 mA).
Default ALM: conducts healthy, opens on fault.
HIGH / unplugged: latch fault; disable all drivers.
No 5 V / 24 V connection to ALM or PB1.

System wiring and linked perfboard assembly map | all voltages DC
ProtoMate Control

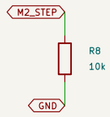
Sheet: /M1 – DM542T/
File: M1_DM542T.kicad_sch

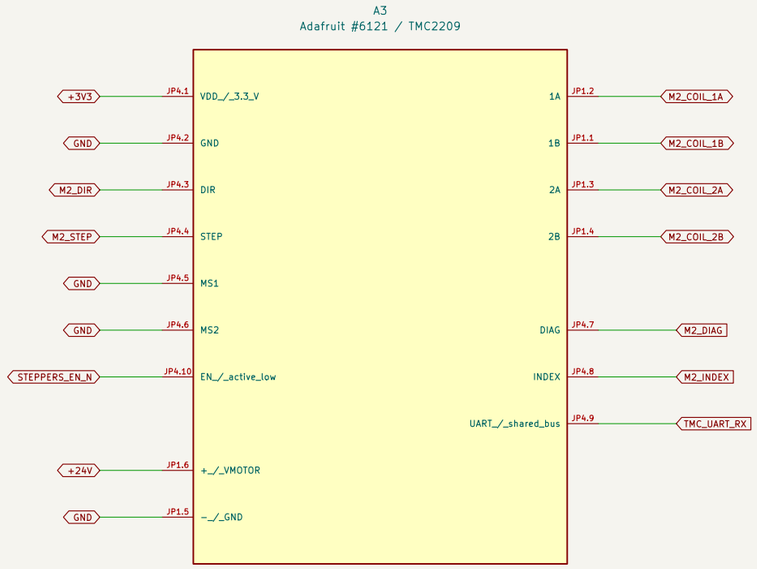
Title: ProtoMate SCARA – M1 – DM542T

Size: A3	Date: 2026–10–15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 2/6

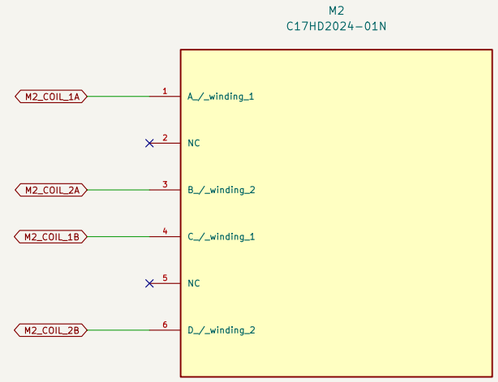
M2 AND M3 — TMC2209

UART addresses: M2 MS1/MS2 = GND/GND (0); M3 = 3.3 V/GND (1). SPRD jumper OPEN.
Firmware selects 1/16 step and SpreadCycle; digital current bypasses the potentiometers.

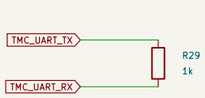




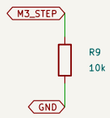


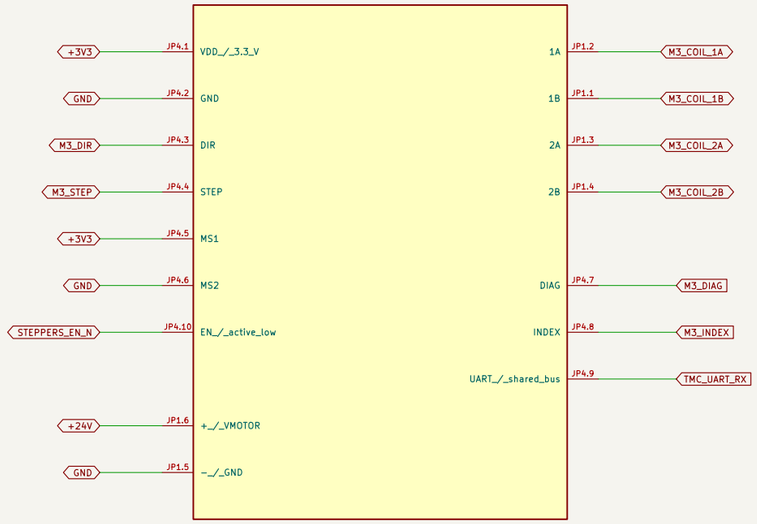


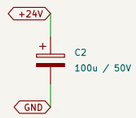
Perfboard bulk capacitor at power branch.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.

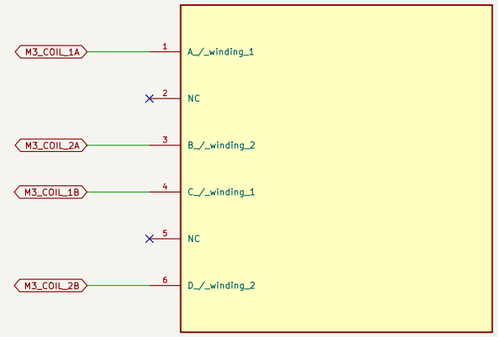


USART2 115200 8N1: TX via R29; RX directly on shared UART.
DIAG: separate active-high fault inputs; no sensorless homing.









Perfboard bulk capacitor at power branch.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.

M2: 1.0 A/phase; UART run/hold 650 mA RMS requested (625 mA quantized).
M3: 0.84 A/phase; UART run/hold 550 mA RMS requested (511 mA quantized).
24 V motor supply: 3.3 V logic/UART. Drivers remain disabled until both answer and verify.
A3/A4: JP4 + JP1.6/5 solder into J105/J106. JP1.4–1: four-way motor terminals.

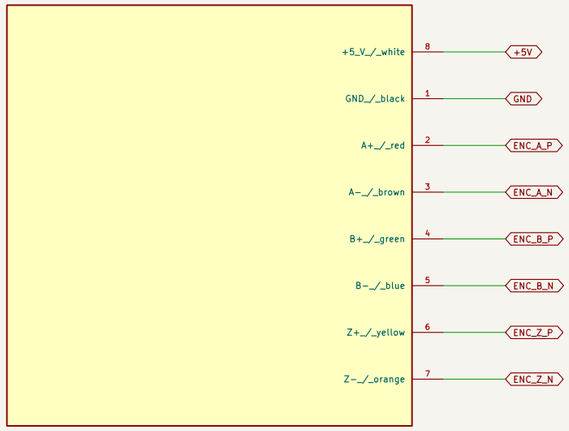
System wiring and linked perfboard assembly map | all voltages DC
ProtoMate Control

Sheet: /M2 and M3 — TMC2209/		
File: M2_M3_TMC2209.kicad_sch		
Title: ProtoMate SCARA — M2 and M3 — TMC2209		
Size: A3	Date: 2026-10-15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 3/6

M1 — ENCODER / HC126 + NPN LEVEL SHIFTERS

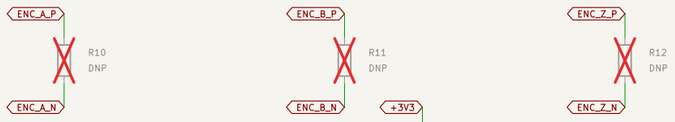
U2 input pairs are deliberately reversed.
Q4–Q6 cancel that inversion before U3.
Encoder connector pinout stays unchanged.
Never wire U2 outputs directly to U3.

J2
M1 encoder / 8-pin

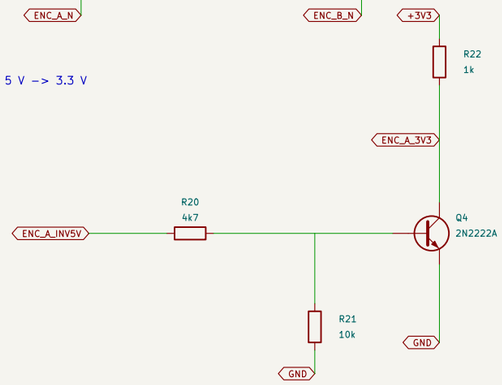


400 P/R -> TIM3 x4 = 1600 counts/rev; Z = once/rev.
Cable colours / connector orientation: motor manual.
Shielded twisted pairs; chassis bond at cable entry.

R10–R12: OPTIONAL TERMINATION, DNP.
Choose for cable and 20 mA encoder output limit.
A+/A−, B+/B−, Z+/Z−: twisted pairs.



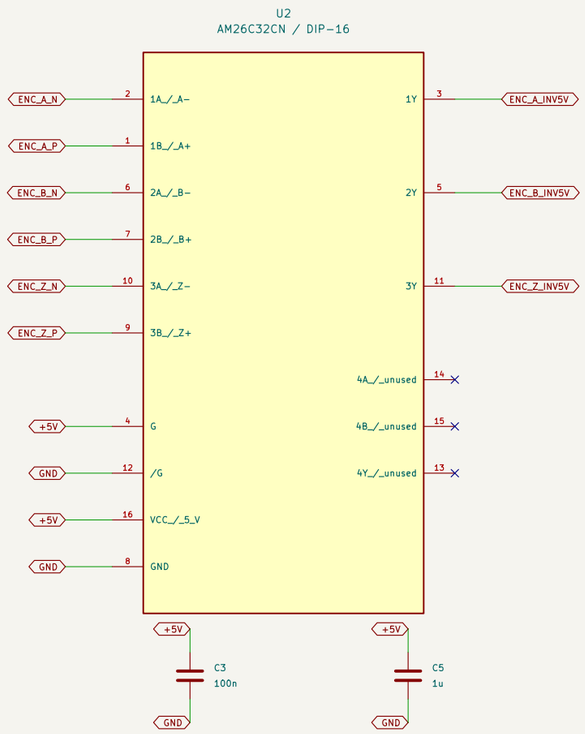
A: 5 V -> 3.3 V



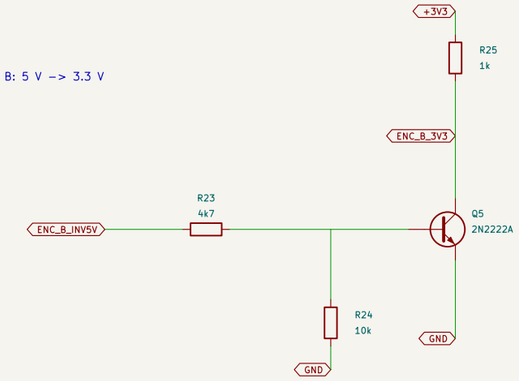
A/B -> D7/PB4 and D5/PA7; Z -> D53/PG7 rising-edge EXTI. All original signal polarities preserved.
Q4–Q6: Diotec 2N2222A E=1/B=2/C=3. Keep collector-to-U3 wires short.
Check edge shape and transistor turn-off delay at the intended encoder speed.

U2: 5 V. Q4–Q6 translate to 3.3 V. U3: 3.3 V ONLY.
U3 OE1/OE2/OE3 HIGH: unused OE4 and A4 LOW.
U2 unused receiver inputs may float (fail-safe).
100 nF at each IC; 1 uF nearby.

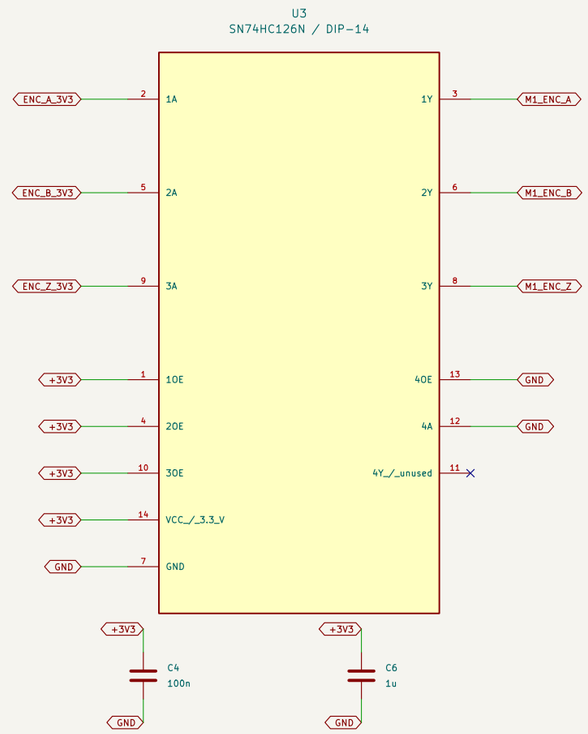
U2
AM26C32CN / DIP-16



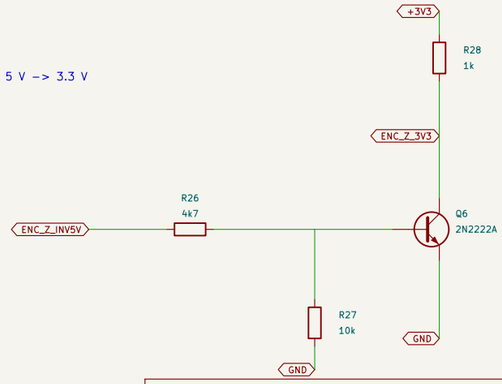
B: 5 V -> 3.3 V



U3
SN74HC126N / DIP-14



Z: 5 V -> 3.3 V



System wiring and linked perfboard assembly map | all voltages DC
ProtoMate Control

Sheet: /M1 — encoder interface/
File: M1_Encoder.kicad_sch

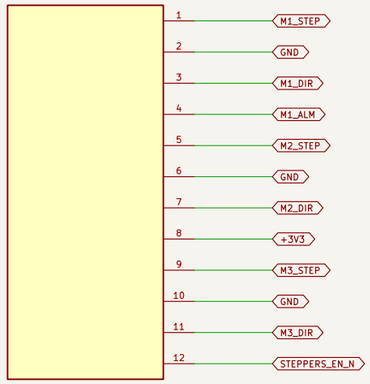
Title: ProtoMate SCARA — M1 — encoder interface

Size: A3	Date: 2026-10-15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 4/6

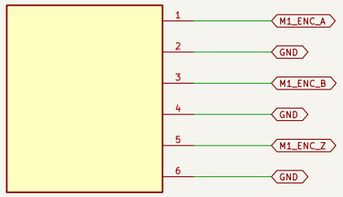
PERFBOARD HEADERS / MODULE MOUNTS

Pin numbers match the layout and Harness.csv. Contacts use the 2.54 mm grid; J103 pairs and J104 terminals span 5.08 mm. Mount pins and cable pins are shown.

J101
GIGA motion / 2x06



J102
GIGA encoder / 2x03

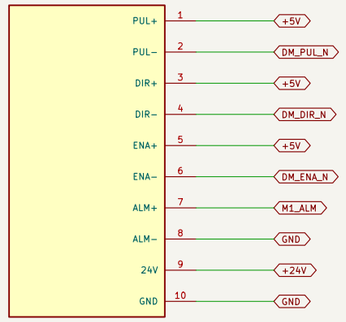


J103
U1 buck direct mount



Pins 3/4: VIN− / GND and VIN+ / 24 V.
24 V enters U1 through its barrel jack.
Pins 1/2: regulated OUT+ / OUT−.

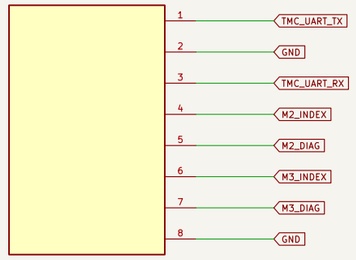
J104
DM542T control + ALM + power / screw 5.08 mm



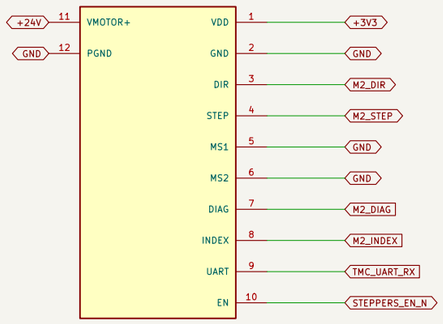
J101: STEP/GND pairs 1/2, 5/6, 9/10.
J101.4 = M1_ALM; .8 = 3.3 V; .12 = EN.
J102 retains A/B/Z ground pairs. J113: GND 2/8.
J103: U1 mount; OUT+/OUT− feed 5 V/GND.
J104: 1–6 control; 7/8 ALM; 9/10 24 V/GND.

J104: 1 PUL+, 2 PUL−, 3 DIR+, 4 DIR−, 5 ENA+, 6 ENA−.
7 ALM+, 8 ALM−, 9 +24 V, 10 GND.
J105/J106: 1 VDD, 2 GND, 3 DIR, 4 STEP, 5 MS1, 6 MS2,
7 DIAG, 8 INDEX, 9 UART shared bus, 10 EN.
11 = JP1.6 / +24 V; 12 = JP1.5 / GND.
J107 follows the numbered J2 encoder connector.

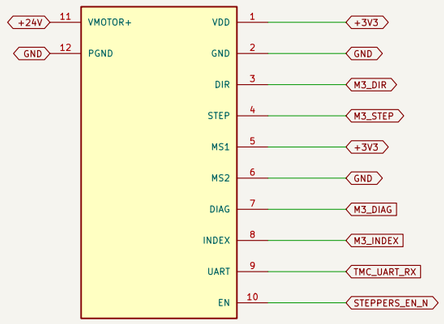
J113
GIGA UART / DIAG / INDEX / 2x04



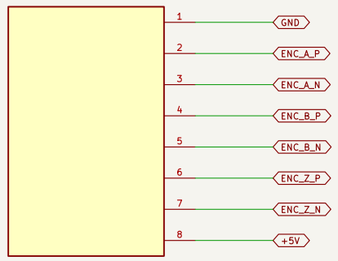
J105
M2 TMC2209 direct mount



J106
M3 TMC2209 direct mount



J107
M1 encoder / 1x08



J113 -> GIGA: 1->D18 (PD5 TX), 3->D19 (PD6 RX),
5->A4 (PC3 M2 DIAG), 7->A6 (PC0 M3 DIAG),
4->D8 (PB8 M2 INDEX), 6->D9 (PB9 M3 INDEX).
Ground contacts: 2 and 8. TX resistor is R29.
INDEX: normal electrical cycle; both-edge EXTI; 3.3 V.

J101 -> GIGA: 1->A7, 3->D4, 5->D3, 7->D6, 9->D2, 11->D16, 12->D15.
J101.4 -> A3 (ALM); .8 -> 3V3; .2/.6/.10 -> common GND.
J103/J105/J106 are direct module mounts, no cable. Other headers use cables. Matching names are one conductor.
J108–J111 (GIGA reference inputs and the three NC switches): see the Reference limit switches sheet.

J101–J107 and J113 mount on the perfboard; power and signals use global names

ProtoMate Control

Sheet: /Perfboard cable headers/
File: Perfboard_Connectors.kicad_sch

Title: ProtoMate SCARA – Perfboard cable headers

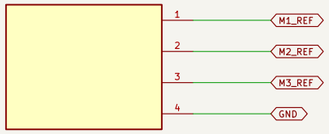
Size: A3	Date: 2026-10-15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 5/6

REFERENCE LIMIT SWITCHES — NORMALLY CLOSED

J108–J111 mount on the perfboard. SW1–SW3 mount on the robot. Global labels connect the cable endpoints.

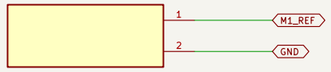
Switches shown released (COM–NC closed).

J108
GIGA references / 1x04



J108 –> GIGA A1
1 = D49 / PE4 (M1)
2 = D51 / PE5 (M2)
3 = D47 / PB2 (M3)
4 = shared GND

J109
M1 NC switch / 1x02

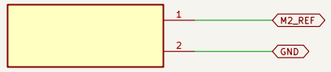


J109.1 –> verified NC contact
J109.2 –> verified COM contact
NO contact stays unconnected.

SW1
M1 Ender–3 V2 / NC



J110
M2 NC switch / 1x02



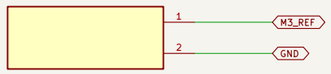
J110.1 –> verified NC contact
J110.2 –> verified COM contact
NO contact stays unconnected.

SW2
M2 Ender–3 V2 / NC



INPUT BEHAVIOR
GIGA internal pull–ups to 3.3 V.
Released: COM–NC closed –> LOW.
Pressed / unplugged: open –> HIGH.
Both–edge EXTI; raw, un–debounced inputs.
M4 firmware handles reference and motion–stop behavior.

J111
M3 NC switch / 1x02



J111.1 –> verified NC contact
J111.2 –> verified COM contact
NO contact stays unconnected.

SW3
M3 Ender–3 V2 / NC



BEFORE WIRING: With the switch disconnected, verify continuity: closed when released, open when pressed.
COM / NC / NO identify contact functions, NOT the module connector pin order. Do not infer power wiring from V/G/S printing.
Use the unpopulated mechanical version as a dry contact; no supply connection. Route each REF with its GND return, away from motor wiring.

NC dry contacts | D49/PE4, D51/PE5, D47/PB2 | 3.3 V pull–ups
ProtoMate Control

Sheet: /Reference limit switches/
File: Reference_Switches.kicad_sch

Title: ProtoMate SCARA — Reference limit switches

Size: A3	Date: 2026–10–15	**Rev: 6.0**
KiCad E.D.A. 10.0.6		Id: 6/6